

# Autonomous and Mobile Robotics

Lecture Notes

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# Preface

These notes were written during the *Autonomous and Mobile Robotics* course held by Prof. Giuseppe Oriolo at Sapienza University of Rome. The main purpose of this document is to provide a coherent and compact reference for the topics covered: kinematics of mobile robots, motion planning, motion control, localization, mapping, SLAM, and decision making under uncertainty. Theoretical derivations are interspersed with practical remarks, formulas, and, where useful, pseudocode or short implementations.

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# Chapter 1

## Introduction: Exploration and Exploitation

**Definition 1.1** (Reinforcement Learning). *Reinforcement learning is the process of learning what to do in order to maximize a numerical reward signal.*

The learner must discover which actions yield the greatest reward by trying them. A key issue is that immediate reward alone is not sufficient: subsequent rewards may occur and may drastically affect the regret if they are not taken into account. Hence, an action may affect not only the immediate reward, which characterizes reinforcement learning as a trial-and-error process.

### 1.1 Exploitation versus Exploration

The central trade-off in reinforcement learning is exploration versus exploitation. Exploration aims to expand the agent's knowledge so that better action selections can be made in the future. Exploitation uses the current knowledge to choose the action that maximizes reward and minimizes regret.

One of the main problems in reinforcement learning is deciding when to continue exploring, that is, collecting more information about the environment, and when to exploit, that is, using the available information to move closer to the goal. Exploration must be carried out carefully and systematically. The **multi-armed bandit** (MAB) problem is a simplified problem that captures the exploration paradigm.

#### 1.1.1 Multi-Armed Bandit

A  $K$ -armed stochastic bandit problem<sup>1</sup> is defined by a collection of  $K$  probability distributions  $\nu_1, \dots, \nu_K$ <sup>2</sup> supported on the interval  $[0, 1]$ , with unknown expected values

$$\mu(a) = \mathbb{E}_{r \sim \nu_a}[r]$$

for each action  $a$ <sup>3</sup> in the set of actions  $\mathcal{A} = \{1, \dots, K\}$ <sup>4</sup>. Here,  $\mu(a)$ <sup>5</sup> denotes the expected reward, and  $\mathbb{E}$ <sup>6</sup> is the expectation operator. The reward  $r$ <sup>7</sup> is drawn from the corresponding distribution.

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<sup>1</sup>  $K$  is the number of arms (actions) available.

<sup>2</sup>  $\nu_a$  is the probability distribution of rewards for arm  $a$ .

<sup>3</sup>  $a$  denotes an action, that is, an arm that can be selected.

<sup>4</sup>  $\mathcal{A}$  is the set of available actions.

<sup>5</sup>  $\mu(a)$  is the expected value (mean) of the reward for arm  $a$ .

<sup>6</sup>  $\mathbb{E}$  denotes the expected value, that is, the theoretical mean.

<sup>7</sup>  $r$  is the reward obtained.

The interaction between the agent and the environment occurs sequentially over a known time horizon of  $T$  rounds<sup>8</sup>:

1. At each round  $t = 0, 1, \dots, T-1$ <sup>9</sup>, the agent selects an arm  $A_t \in \{1, \dots, K\}$ <sup>10</sup> based on the information observed up to the previous round.
2. The environment independently draws a reward  $r_t \sim \nu_{A_t}$ <sup>11</sup>, with expected value  $\mathbb{E}[r_t | A_t] = \mu(A_t)$ , and reveals only  $r_t$  to the agent.

We define an optimal arm as  $a^* \in \arg \max_{a \in \mathcal{A}} \mu(a)$ <sup>12</sup>, with maximum expected value  $\mu^* = \mu(a^*)$ <sup>13</sup>. For each arm  $a$ , the suboptimality gap  $\Delta(a)$ <sup>14</sup> is

$$\Delta(a) = \mu^* - \mu(a). \quad (1.1)$$

**Notation.** The expected reward of an arm is written as a function  $\mu(a)$  rather than with a subscript  $\mu_a$  or  $\mu_k$ . Both notations are equivalent, since the set of arms is  $\mathcal{A} = \{1, \dots, K\}$ . The functional form  $\mu(a)$  is preferred for several reasons. First, at round  $t$  the agent selects a random arm  $A_t$ , and writing the expected reward as  $\mu(A_t)$  is more natural than  $\mu_{A_t}$ . Second, the same functional notation is used later for empirical estimates  $\hat{\mu}_t(a)$  and for counts  $N_t(a)$ . Third, in contextual bandits the expected reward becomes a function of both context  $z$  and action  $a$ , i.e.  $r(z, a)$ , and in reinforcement learning it becomes  $r(s, a)$  or  $Q(s, a)$ . Using  $\mu(a)$  from the beginning prepares for these generalisations. Finally, if the action space were continuous, a subscript  $k$  would not be possible, whereas  $\mu(a)$  remains valid.

### 1.1.2 Cumulative Regret and Decomposition

The agent's objective is to minimize the cumulative regret  $\text{Reg}(T)$ <sup>15</sup> over  $T$  rounds, which measures the performance loss caused by not always selecting the optimal arm:

$$\text{Reg}(T) = T\mu^* - \sum_{t=0}^{T-1} \mu(A_t). \quad (1.2)$$

By the regret decomposition lemma, the total loss can be expressed in terms of the number of pulls

$$N_T(a) = \sum_{t=0}^{T-1} \mathbf{1}[A_t = a]$$

of each arm  $a$ . Here,  $N_T(a)$ <sup>16</sup> counts how many times arm  $a$  was selected, and  $\mathbf{1}[\cdot]$ <sup>17</sup> is the indicator function. Therefore,

$$\text{Reg}(T) = \sum_{a=1}^K \Delta(a) N_T(a). \quad (1.3)$$

<sup>8</sup> $T$  is the total number of rounds (time horizon).

<sup>9</sup> $t$  denotes the current round.

<sup>10</sup> $A_t$  is the arm chosen by the agent at round  $t$ .

<sup>11</sup> $r_t$  is the reward obtained at round  $t$ .

<sup>12</sup> $a^*$  is an optimal arm, that is, the one with the highest mean.

<sup>13</sup> $\mu^*$  is the expected value of the optimal arm.

<sup>14</sup> $\Delta(a)$  is the difference between the optimal mean reward and the mean reward of arm  $a$ .

<sup>15</sup> $\text{Reg}(T)$  is the cumulative regret, that is, the total loss compared to the optimal arm.

<sup>16</sup> $N_T(a)$  is the number of times arm  $a$  has been selected up to round  $T$ .

<sup>17</sup> $\mathbf{1}[\cdot]$  equals 1 if the condition is true, and 0 otherwise.

**Interpretation of the decomposition.** The gap  $\Delta(a) = \mu^* - \mu(a)$  is not a random reward; it is a fixed quantity determined by the true, unknown expected values  $\mu(a)$ . Therefore, every single time arm  $a$  is pulled, the expected loss is exactly  $\Delta(a)$ . If arm  $a$  is pulled  $N_T(a)$  times, the total expected loss caused by that arm is the sum of  $N_T(a)$  identical terms, i.e.  $\Delta(a) \cdot N_T(a)$ . Summing over all arms gives the total regret. The formula can be written explicitly as

$$\text{Reg}(T) = \underbrace{\Delta(1)N_T(1)}_{\text{loss from arm 1}} + \underbrace{\Delta(2)N_T(2)}_{\text{loss from arm 2}} + \cdots + \underbrace{\Delta(K)N_T(K)}_{\text{loss from arm } K}.$$

Thus each arm has its own gap and its own count; the decomposition does not assume that all arms have the same regret. It simply groups the time-wise sum  $\sum_{t=0}^{T-1} \Delta(A_t)$  by arm.

**Numerical example.** Consider  $K = 3$  arms and  $T = 10$  rounds. Let the true means be  $\mu(1) = 0.8$  (optimal),  $\mu(2) = 0.5$ , and  $\mu(3) = 0.1$ . Then  $\mu^* = 0.8$ , and the gaps are

$$\Delta(1) = 0.8 - 0.8 = 0, \quad \Delta(2) = 0.8 - 0.5 = 0.3, \quad \Delta(3) = 0.8 - 0.1 = 0.7.$$

Suppose the algorithm selects arm 1 five times, arm 2 three times, and arm 3 two times. Then

$$\text{Reg}(10) = 0 \cdot 5 + 0.3 \cdot 3 + 0.7 \cdot 2 = 0 + 0.9 + 1.4 = 2.3.$$

This illustrates that arms with larger gaps contribute more to the total regret per pull.

### 1.1.3 Algorithms and the Exploration–Exploitation Balance

- **Greedy rule:** Select each arm once and then commit to the arm with the highest empirical mean. This strategy suffers linear regret  $\Omega(T)$ .
- **Explore-then-commit (ETC):** Explore each arm uniformly for  $m$  rounds<sup>18</sup> and subsequently use only the empirically best arm. With an optimal choice of  $m^*$ , it guarantees regret of order

$$\mathcal{O}\left(T^{2/3} K^{1/3} \ln^{1/3}(2K/\delta)\right),$$

where  $\delta$ <sup>19</sup> is a confidence parameter.

- **Upper confidence bound (UCB):** Apply the principle of optimism in the face of uncertainty by adding a confidence interval  $b_t(a)$ <sup>20</sup> to the empirical mean  $\hat{\mu}_t(a)$ <sup>21</sup>. It guarantees near-optimal regret

$$\tilde{\mathcal{O}}(\sqrt{KT}),$$

where  $\tilde{\mathcal{O}}$ <sup>22</sup> hides logarithmic factors.

<sup>18</sup> $m$  is the number of exploration rounds per arm in the ETC strategy.

<sup>19</sup> $\delta$  is a confidence parameter (error probability).

<sup>20</sup> $b_t(a)$  is the width of the confidence interval added to the empirical mean.

<sup>21</sup> $\hat{\mu}_t(a)$  is the empirical mean of arm  $a$  observed up to round  $t$ .

<sup>22</sup> $\tilde{\mathcal{O}}$  denotes an asymptotic bound that ignores logarithmic factors.